Lab 3

Variant D
Group 34
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1. Introduction

a. Task description

The task of this laboratory is to minimise the 2D Styblinski-Tang function using a genetic algorithm. The function is $f(x1, x2) = 0.5 * (x1^4 - 16 x1^2 + 5 x1 + x2^4 - 16 x2^2 + 5 x2)$ and we search it on the domain [-5, 5] x [-5, 5]. It has four local basins and a single global minimum at (-2.903, -2.903) with $f^* = -78.332$. The global basin is clearly the deepest one, so a well-tuned GA should find it most of the time, while the other three act as traps.

b. Algorithm description

A genetic algorithm works with a population of candidate solutions encoded as binary strings. In every generation we evaluate the fitness of each individual, select parents by tournament, combine them with one-point crossover and apply per-gene bit-flip mutation. Over time the population drifts towards the region of the search space with the best fitness. For a minimisation problem like ours the winner of a tournament is the individual with the smallest objective value.

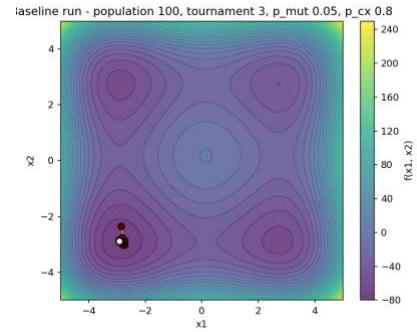
2. Implementation

We implemented all required methods on top of the provided skeleton. Each variable is encoded with 20 bits, so the full chromosome has 40 genes and the resolution along each axis is about $10 / (2^{20} - 1) \sim 1e-5$, which is more than enough for this function. Decoding uses the min-max normalisation helper from the template. The mutation operator flips every gene independently with the chosen probability; crossover picks a random split point and swaps the tails of the two parents. Tournament selection picks the individual with the smallest f among k randomly chosen candidates. The main loop records the best individual of every generation so we can draw the trace plot at the end.

Baseline settings, taken from the template: population = 100, tournament size = 3, p_mutation = 0.05, p_crossover = 0.8, 50 generations. With those values the algorithm reaches the global minimum essentially every run. The figure below shows one baseline run, with the best individual of each generation drawn as a dot going from dark red (generation 0) to light red (generation 49).

3. Discussion

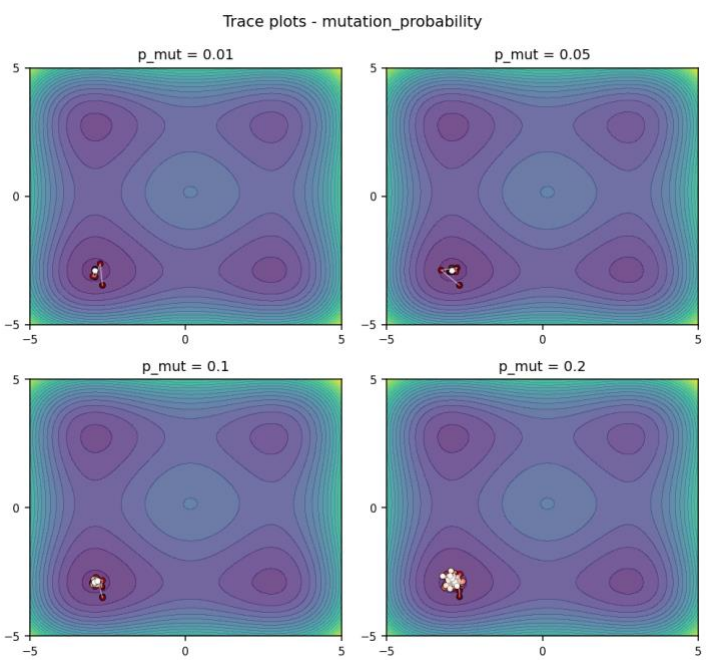
For the parameter study we ran the GA 10 times per configuration and varied only one parameter at a time while keeping the rest at the baseline. Below we report the mean of the final f-value across the 10 runs (the global minimum is -78.332).



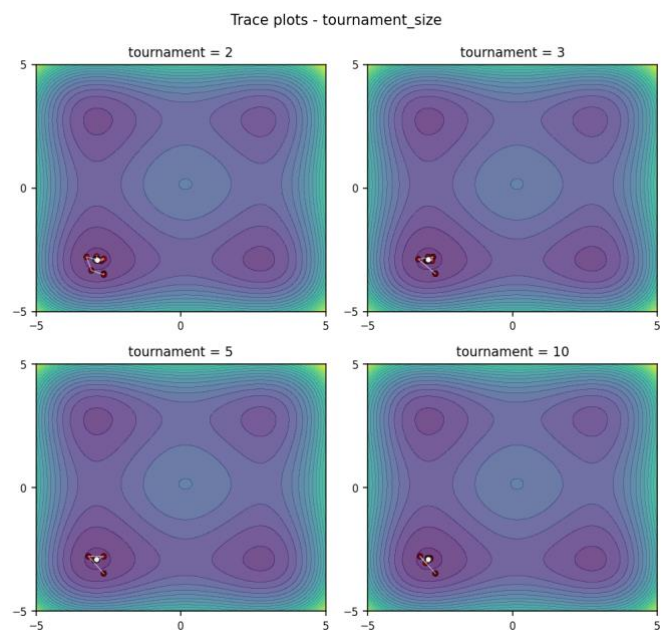
Baseline run - trace of the best individual per generation.

Mutation probability (0.01, 0.05, 0.1, 0.2). Values up to 0.05 all reach the global minimum with a standard deviation close to zero. At 0.1 the mean drops slightly to -78.30 because mutation occasionally flips useful bits of the best individual. At 0.2 the population is basically randomised every generation, the mean jumps to -77.90 and the std goes up to ~0.58. The sweet spot is clearly around 0.01 - 0.05.

Tournament size (2, 3, 5, 10). Size 2 gives weak selection pressure and a few runs do not fully converge (mean -78.32, std ~0.02). From size 3 onwards the algorithm converges essentially perfectly. We did not see a clear over-exploitation effect with size 10, probably because mutation keeps the population diverse enough.

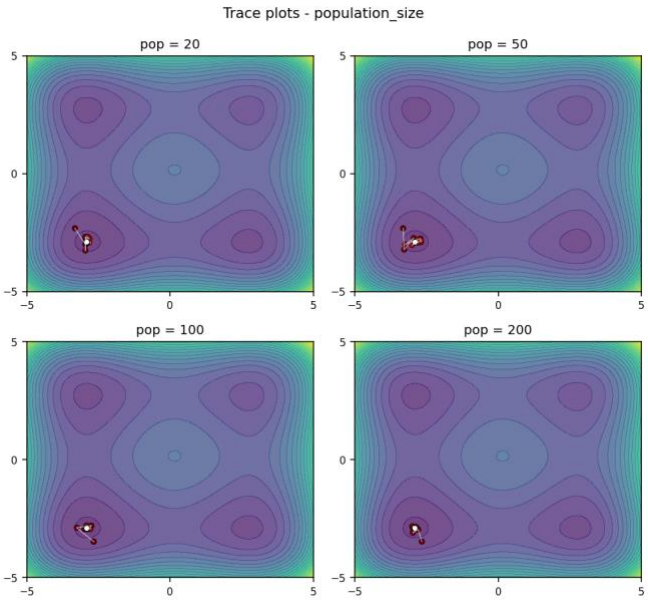


Trace of the best individual per generation for each mutation probability value.

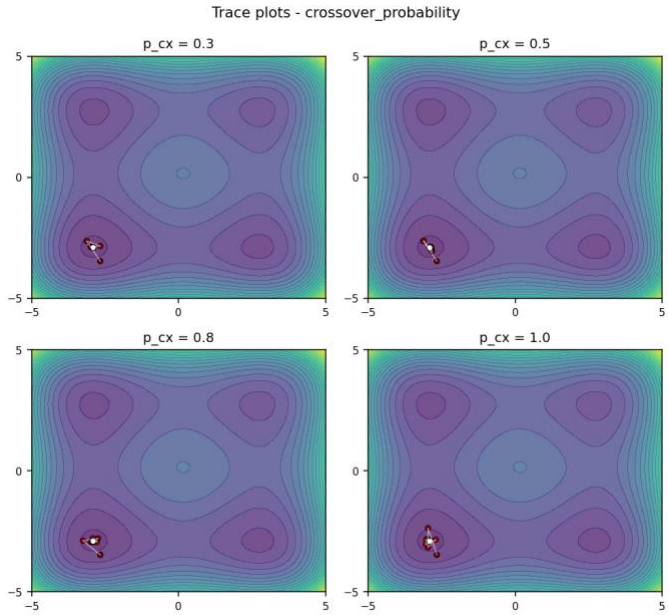


Trace of the best individual per generation for each tournament size.

Population size (20, 50, 100, 200). With only 20 individuals the initial sampling is too sparse and one of our runs got stuck at about -75.9 (std ~0.73). From 50 upwards every run reaches the global minimum. 200 is marginally more precise than 100 but noticeably slower per generation, so 100 is the best trade-off.



Trace of the best individual per generation for each population size.



Trace of the best individual per generation for each crossover probability.

Crossover probability (0.3, 0.5, 0.8, 1.0). This parameter had the smallest effect. 0.3, 0.5 and 1.0 all converge to -78.332 with std ~0. The value 0.8 shows a larger std only because one of the ten runs happened to converge to a local minimum; with only 10 repetitions this is a variance artifact rather than a real effect of that probability.

4. Conclusion

For Styblinski-Tang in 2D the GA is fairly forgiving. The parameters that actually matter are mutation probability and population size: mutation around 0.05 and at least ~50 individuals are enough to reach the global minimum every time. Tournament size only matters if it is too small (size 2 is a bit weak); crossover probability barely matters in the 0.3 - 1.0 range. A reasonable default is therefore population ~100, tournament size 3, p_mut = 0.05 and p_cx = 0.8 - which is also what the template suggested. The GA normally reaches f* = -78.332 within the first 10 - 20 generations and spends the remaining generations on the last decimals.